POAMS: LECTURE 18
ANGULAR MOMENTUM AND SPIN

In this lecture, we study the incorporation of spin into orbital angular momentum in POAMS. In Viv and I's approach, it is not just orbital angular momentum which is holistically conserved but, rather, the total angular momentum of a particle, i.e. orbital plus spin. This commits us to predictions that are distinct from those of Newtonian theory. We begin by studying the spin angular momentum and spin kinetic energy of a particle. We continue by showing how the spin of an orbiting particle in the two-body problem is incorporated in its orbital motion by applying conservation of the total angular momentum. This lecture is based on some of Chapter 7 of 'Light-speed....'.

1. Introduction

Over the last four lectures, I have presented Viv and I's alternative approach to Newton's theory of gravitation, known as the *Pope-Osborne Angular Momentum Synthesis* (POAMS). Recall that POAMS begins with the premise that the observable universe is in *holistic* balance, so that the *natural* or unrestricted motion of any body is instantly correlated with that of each and every other body within the system. Recall also that natural, force-free motion in POAMS is not rectilinear as in Newtonian theory, but is naturally curved and orbital. POAMS postulates that the agency responsible for maintaining this holistic balance is *conservation of angular momentum*. Also, in POAMS, the observable universe is ultimately discrete on the micro-phenomenal level, with angular momentum quantised (following Bohr) in discrete units of $h/2\pi$, where h is *Planck's constant*. POAMS make no distinction between the macro- and micro-phenomenal levels so far as the conservation of angular momentum is concerned. [*End of recap*]

So far, we have considered only the orbital angular momentum of a particle in relation to some fixed point of reference, particularly in relation to the centre-of-mass of some larger body. However, there is another, and in some sense more fundamental, angular momentum associated with any particle. This is the *intrinsic* or *spin* angular momentum, associated with a spinning particle. In POAMS, therefore, it is not just orbital angular momentum which is holistically conserved but, rather, the *total* angular

momentum of a particle, *i.e. orbital* plus *spin*. This commits us to predictions that are distinct from those of Newtonian theory. In particular, POAMS predicts certain relatively small but nevertheless measurable effects associated with macroscopic spinning bodies. In contrast, on the micro-phenomenal scale, the proportion of spin to orbital angular momentum is much larger. We show that by incorporating these spin angular momenta into the 'orbital' parameters of the 'electron' and 'proton' in the Bohr atom, it is possible to obtain the accepted physical constants associated with the hydrogen atom without having to refer to the customary 'electrostatic forces' of Bohr's derivation, provided in Lecture 10. The POAMS approach thus indicates that angular momentum consideration leads to the logical redundancy, not only of *in vacuo* 'gravitational' forces but also of 'electrostatic forces'.

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2. Spin angular momentum

In Lecture 13, we defined the **angular momentum**, **L**, of a particle P of mass m relative to a chosen origin, O, as

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} \quad (1)$$

[*Lecture 13, equation (2)*] where $\mathbf{r}(t)$ is the position vector of P relative to O, and $\mathbf{v}(t) = \mathrm{d}\mathbf{r}/\mathrm{d}t$ is the velocity of P. Remember that **L** is a vector that lies perpendicular to the plane containing **r** and **v**. In order to distinguish **L** from spin angular momentum, from now on we shall refer to **L** as the **orbital angular momentum** of P relative to O. We showed in Lecture 13 that if the trajectory of P relative to O lies in a plane, then

$$L = \|\mathbf{L}\| = mr^2\omega$$

[*Lecture 13, equation (7)*] where $r = \|\mathbf{r}\|$ and ω is the *angular speed* of P. Here, the quantity mr^2, which gives information about both the matter content and the shape of the orbit of P, is known as the *moment of inertia*, I, of P. Then

$$L = mr^2\omega = I\omega \quad (2)$$

In the same way, the magnitude of the **spin angular momentum**, **S**, of a rotating particle depends on the particle's moment of inertia and its angular speed. In general, these measures are very involved and here we shall deal only with the case of an axially symmetric body spinning about a fixed axis, which will be sufficient for the practical examples that follow. These examples will give an indication of why, according to POAMS, spin has an automatic and immediate effect on orbital motion in general.

Let us consider, then, a rigid *axially-symmetric body*, *P*, which is spinning (rotating) about its own fixed axis of symmetry. Examples of such a body include (i) a sphere rotating about an axis through its centre, (ii) a circular disc spinning about an axis perpendicular to the disc and passing through the centre of the disc, and (iii) a cube rotating about any axis passing through its centre and the centre of any of its faces. In such cases, the **magnitude**, *S*, of *P*'s **spin or rotational angular momentum**, (about the axis of rotation) is defined by

$$S = I\omega \quad (3)$$

where *I* is *the moment of inertia* of *P* and ω is the angular speed of *P* about the axis of rotation. (Compare this with equation (2).) The moment of inertia, more accurately called the *rotational inertia*, of a rigid particle is a quantity that determines the torque required to produce a desired angular acceleration about the fixed rotational axis. In general, for a particle of mass *M* it is given by

$$I = \int_0^M r^2 \, \mathrm{d}m$$

where d*m* is the mass element of the particle and *r* is the distance of any point in or on the body from the axis of rotation (compare with equation (2)). As in the case of orbital motion, *I* depends on the mass of *P* and its shape. In general, the derivation of the moment of inertia for a particular shape using this definition can be very involved. For our purposes, it is sufficient to consider two types of rigid axially-symmetric bodies and for these, we quote *I* without derivation. For the mass enclosed by a *sphere of radius r*, about its axis of rotation,

$$I = \frac{2}{5}mr^2 \quad (4)$$

[*see endnote (**) for details*] and for a *circular disc of radius r*, about its axis of rotation,

$$I = \frac{1}{2}mr^2 \qquad (5)$$

[*see endnote (*) for details*]. In any case, the **spin or rotational angular momentum** of *P* is then the vector **S** defined by **S** = *S***n**, where **n** is a vector of unit length in the direction of the axis of rotation.

Because the moment of inertia of even an axially-symmetric body depends on its shape, it is often more convenient to express the spin angular momentum of such a body in terms of its *rotational* or *spin kinetic energy.* Equation (5) of Lecture 13 [$v^2 = \left(\frac{dr}{dt}\right)^2 + r^2\omega^2$] and equation (2) tell us that the kinetic energy, *K*, of a particle of mass *m* in *a circular orbit of radius r*, is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}mr^2\omega^2 = \frac{1}{2}I\omega^2 \qquad (6)$$

where *I* is the moment of inertia of *P*'s orbit. It then follows, also from equation (2), that

$$L = \frac{2K}{\omega} \qquad (7)$$

By analogy with equation (6), in general, the **rotational or spin kinetic energy**, K_S, of a rigid axially-symmetric body, about its axis of rotation is defined by

$$K_S = \frac{1}{2}I\omega^2 \qquad (8)$$

where *I* is the moment of inertia of the body and ω is its angular speed. It then follows immediately from equation (3) that the magnitude of the body's spin angular momentum is

$$S = \frac{2K_S}{\omega} \qquad (9)$$

regardless of the shape of the body.

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3. The incorporation of spin into orbital angular momentum in POAMS

In classical dynamics, Newton's laws are applied to any system containing spinning bodies in orbital motion by treating a spinning body itself as a system of many discrete and self-sufficient orbiting particles. In contrast, in POAMS because of the holistic conservation of angular momentum, in which the motions of all bodies are immediately balanced with one another as a system, any effects due to spin are *instantly correlated with the total angular momentum.* POAMS accounts for this instantaneous correlation by postulating that the kinetic energy of an orbiting spinning particle depends on both its spin kinetic energy and the orbital kinetic energy it would have if it were not spinning. This implies that in orbital systems of constant angular momentum involving spin, the 'gravitational constant' G has to be replaced by a more general function, $\mathcal{G}$, of the orbital parameters.

For simplicity, we once again consider only the case of a freely moving particle, P, of mass m, in a closed orbit about a body of mass M, with $m \ll M$. If P spins on its axis while orbiting, then according to Newtonian dynamics, if the spinning body is perfectly spherical, the spin situation is the same as that of the non-spin. In contrast, according to POAMS, since any orbital angular momentum is balanced against the total momentum of the system, introducing a spin at any one place has a direct and immediate effect on the system as a whole. In other words, POAMS postulates that P's spin will cause it take up a natural orbit different from that which it would follow if it were not spinning.

Suppose for the moment that P does not spin and follows a natural elliptical orbit about the centre-of-mass of B, with radial coordinate $r_0(t)$ and with angular speed $\omega_0(t)$, at any time t. From equation (11) of Lecture 15 $[\frac{d^2r}{dt^2} - \frac{L^2}{m^2r^3} = -\frac{\alpha}{r^2}]$, in POAMS, just as in Newtonian theory, this natural elliptical orbit is determined by

$$\frac{\mathrm{d}^2 r_0}{\mathrm{d}t^2} - \frac{L^2}{m^2 r_0^3} = -\frac{GM}{r_0^2} \qquad (10)$$

where L is the constant magnitude of P's orbital angular momentum. In order to distinguish between orbital and spin kinetic energy, we shall denote P's orbital kinetic energy by K_o. It then follows by equations (5) and (7) of Lecture 13 $\left[v^2 = \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + r^2\omega^2,\ L = mr^2\omega\right]$ that

$$K_o = \frac{1}{2}mv_0^2 = \frac{1}{2}m\left(\left(\frac{\mathrm{d}r_0}{\mathrm{d}t}\right)^2 + \frac{L^2}{m^2 r_0^2}\right) \qquad (11)$$

where $v_0(t)$ is the speed of P in its natural orbit.

Assume now that P is axially symmetric and spins on its axis whilst orbiting the body B. POAMS postulates that the parameters of P's orbit will be directly affected by P's spin. Explicitly, when spinning, P follows a new closed natural orbit with radial coordinate $r_s(t)$ and angular speed $\omega_s(t)$, where, following equation (11), $r_s(t)$ is determined by

$$\mathcal{K} = \frac{1}{2}m\left(\left(\frac{\mathrm{d}r_s}{\mathrm{d}t}\right)^2 + \frac{L^2}{m^2 r_s^2}\right) \qquad (12)$$

where $\mathcal{K}$ is P's new orbital kinetic energy. POAMS postulates that $\mathcal{K}$ depends on both the orbital kinetic energy, K_o, of the orbit which P would naturally follow if it were not spinning, and its spin kinetic energy, K_s. In other words, P's spin kinetic energy is transferred directly to its new orbital kinetic energy, $\mathcal{K}$, such the overall angular momentum is conserved. Once $\mathcal{K}$ is known, $r_s(t)$ may be obtained by solving the first order differential equation (12). It then follows by equation (7) of Lecture 13 $[L = mr^2\omega]$ that

$$\omega_s(t) = \frac{L}{mr_s^2} \qquad (13)$$

Since this 'new' closed orbit of P about B, as determined by its spin, is a natural force-free orbit, it must satisfy an equation of the form (10) in terms of the radial coordinate $r_s(t)$, so that in this case,

$$\frac{\mathrm{d}^2 r_S}{\mathrm{d}t^2} - \frac{L^2}{m^2 r_S^3} = -\mathcal{G}\frac{M}{r_S^2} \quad (14)$$

where $\mathcal{G}$ is a new factor which replaces the 'gravitational constant' G. This factor $\mathcal{G}$ reduces to G when and only when equation (14) describes any natural, spin-less orbit of P and so reduces to equation (10). In other words, $\mathcal{G} = G$ when and only when $\mathcal{K} = K_o$. Once the orbital parameter $r_s(t)$ is known, the corresponding 'gravitational factor' $\mathcal{G}$ may be calculated from equation (14).

For a *circular* orbit of P, this process simplifies considerably. In this case, it follows by the definition of orbital kinetic energy that the speed, v_s, of P in its natural orbit, adjusted for spin (that is, P's *effective orbital speed*), is given by

$$v_s^2 = \frac{2}{m}\mathcal{K} \qquad (15)$$

Then r_s may be calculated using equation (7) of Lecture 16 [$L = mvr$], that is,

$$r_S = \frac{L}{mv_S} \qquad (16)$$

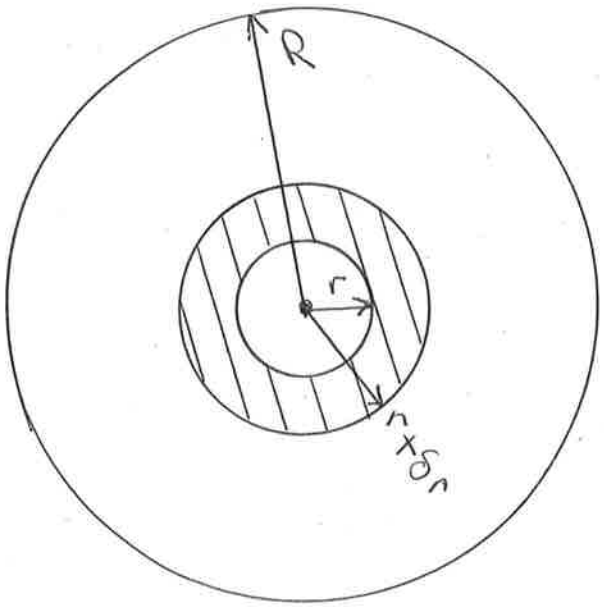
Finally, in this case it follows by equation (8) of Lecture 16 [$v^2 = \frac{GM}{r}$] that equation (14) reduces to

$$\mathcal{G} = \frac{r_S v_S^2}{M} \qquad (17)$$

In the next lecture, I will continue to discuss, in Normal Realist terms, the effect of spin on the orbital parameters of free-moving particles and hence, on the value of 'G'. In order to illustrate these ideas, I will present a numerical example.

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(*) The moment of inertia, I, of a <u>circular disc</u> of radius R and mass M about an axis through its centre and orthogonal to the plane of the disc.



Let a small 'ring' element of the disc have inner radius r and outer radius $r+\delta r$.

Then the area element of the disc is

$$\delta A \cong 2\pi r\,\delta r$$

The mass per unit area of the disc $= \dfrac{M}{\pi R^2}$, so that δm, the mass element of the disc is

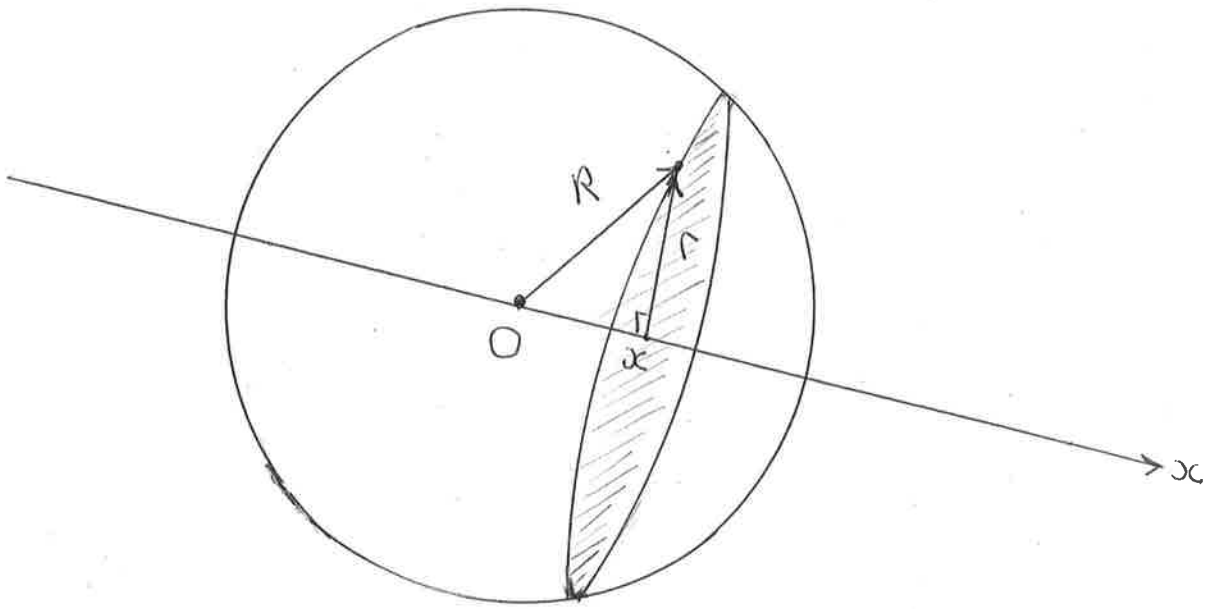
$$\delta m \cong \frac{M}{\pi R^2} 2\pi r\,\delta r.$$

Hence, $$I = \int_0^R r^2\,dm = \int_0^R r^2 \left(\frac{M}{\pi R^2} 2\pi r\right) dr$$

$$\therefore I = \frac{2M}{R^2}\int_0^R r^3\,dr = \frac{2M}{R^2}\left[\frac{r^4}{4}\right]_0^R = \underline{\underline{\frac{1}{2}MR^2}}$$

(Equation (5))

(**) The moment of inertia, I, of a sphere of radius R and mass M (i.e. the mass enclosed by the sphere) about an axis through its centre.



Consider a disc of thickness δx, orthogonal to the axis of the sphere, with centre distance x from the centre of the sphere, and radius r as shown.

The moment of inertia of the disc is '$\frac{1}{2}MR^2$' (see (*)) so $dI = \frac{1}{2}r^2\,dm$, where $dm = \rho\,dV$. Here, ρ is the density of a thin disc of mass m and dV is its volume element, so that $dV = \pi r^2 dx$ (see diag.)

Hence, $dI = \frac{1}{2}r^2 \rho \pi r^2 dx = \frac{1}{2}\pi r^4 \rho\, dx$

Also, from the diagram, $R^2 = x^2 + r^2$, so $r^2 = R^2 - x^2$.

Hence, $I = \frac{1}{2}\pi\rho \int_{-R}^{R} (R^2 - x^2)^2\, dx$

[When $x = 0$, $R^2 = r^2$, so $R = \pm r$, when $r = 0$, $x = \pm R$]

$$\therefore I = \frac{1}{2}\pi\rho\left[R^4 x - \frac{2R^2x^3}{3} + \frac{x^5}{5}\right]_{-R}^{R}$$

$$= \pi\rho\left(R^5 - \frac{2R^5}{3} + \frac{R^5}{5}\right)$$

$$\therefore I = \pi\rho\left(\frac{8R^5}{15}\right).$$

Also, if V is the volume enclosed by the sphere,

$V = \frac{4}{3}\pi R^3$, so that,

$$\rho = \frac{M}{V} = \frac{3M}{4\pi R^3}.$$

Then $I = \left(\frac{8\pi R^5}{15}\right)\left(\frac{3M}{4\pi R^3}\right) = \frac{2}{5}MR^2$

(Equation (4))